

Euler - Cauchy Equation (Reducible to constant coefficients):

$$a_n x^n D^n y + a_{n-1} x^{n-1} D^{n-1} y + \dots + a_1 x Dy + a_0 y = R(x),$$

Where $a_n, a_{n-1}, \dots, a_1, a_0$ are constants. — (1)

Take $x = e^t \Leftrightarrow t = \ln x$.

$$\frac{dy}{dx} = \frac{dy}{dt} \cdot \frac{dt}{dx} = \frac{dy}{dt} \cdot \frac{1}{x}$$

$$\Rightarrow x \frac{dy}{dx} = \frac{dy}{dt} \Rightarrow \boxed{x Dy = D_1 y}, \quad D_1 = \frac{d}{dt}.$$

$$\frac{d^2 y}{dx^2} = \frac{d}{dx} \left(\frac{dy}{dx} \right) = \frac{d}{dt} \left(\frac{dy}{dx} \right) \frac{dt}{dx}$$

$$= \frac{d}{dt} \left(\frac{1}{x} \frac{dy}{dt} \right) \frac{1}{x} = \frac{d}{dt} \left(e^{-t} \frac{dy}{dt} \right) \frac{1}{x}$$

$$= \frac{1}{x} \left(e^{-t} \frac{d^2 y}{dt^2} - e^{-t} \frac{dy}{dt} \right)$$

$$= \frac{e^{-t}}{x} \left(\frac{d^2 y}{dt^2} - \frac{dy}{dt} \right) = \frac{1}{x^2} (D_1^2 y - D_1 y)$$

$$\Rightarrow \boxed{x^2 D^2 y = D_1(D_1 - 1)y}.$$

By mathematical induction, prove that

$$\boxed{x^n D^n y = D_1(D_1 - 1) \dots (D_1 - (n-1))y}.$$

$\therefore$ If we apply the transformation $x = e^t$, Eq. (1) reduces to constant coefficients diff. eq.

Ex.1 $2x^2 y'' + 3xy' - 3y = x^3$.

Soln Put $x = e^t$, this gives $x D = D_1$,
 $x^2 D^2 = D_1(D_1 - 1), \quad D_1 = \frac{d}{dt}.$

We have $(2x^2 D^2 + 3xD - 3)y = x^3$.

$$\text{Thus } (2D_1(D_1-1) + 3D_1 - 3)y = e^{3t}$$

$$\Rightarrow (2D_1^2 + D_1 - 3)y = e^{3t}.$$

$$\text{Put } y = e^{mt} \text{ in } (2D_1^2 + D_1 - 3)y = 0$$

$$\therefore 2m^2 + m - 3 = 0, \quad m = 1, -3/2.$$

$$\begin{aligned} \therefore \text{C.F.} &= C_1 e^t + C_2 e^{-3/2 t} \\ &= C_1 x + C_2 x^{-3/2} \quad (\because e^t = x). \end{aligned}$$

$$\begin{aligned} \text{PI} &= \frac{1}{2D_1^2 + D_1 - 3} e^{3t} = \frac{1}{2 \times 3^2 + 3 - 3} e^{3t} = \frac{1}{18} e^{3t} \\ &= \frac{1}{18} x^3. \end{aligned}$$

$$\text{General solution } y = C_1 x + C_2 x^{-3/2} + \frac{1}{18} x^3.$$

$$\text{Ex. 2 } x^2 y'' + xy' + y = \tan(\ln x)$$

$$\underline{\text{Soln:}} \quad \text{Put } x = e^t, \quad xD = D_1, \quad x^2 D^2 = D_1(D_1-1), \quad D_1 = \frac{d}{dt}.$$

$$\Rightarrow (D_1(D_1-1) + D_1 + 1)y = \tan t$$

$$\Rightarrow (D_1^2 + 1)y = \tan t.$$

$$\therefore \text{C.F.} = C_1 \underset{y_1}{\cos t} + C_2 \underset{y_2}{\sin t}$$

$$W = \det \begin{bmatrix} \cos t & \sin t \\ -\sin t & \cos t \end{bmatrix} = 1$$

$$\therefore y_p = -\cos t \int \frac{\sin t}{1} \tan t \, dt + \sin t \int \frac{\cos t}{1} \tan t \, dt$$

(by variation of parameters)

$$= -\cos t \int \frac{1 - \cos^2 t}{\cos t} \, dt + \sin t (-\cos t)$$

$$= -\cos t (\ln(\sec t + \tan t) - \sin t) - \sin t \cos t$$

$$= -\cos t \ln(\sec t + \tan t)$$

General soln: $y = c_1 \cos t + c_2 \sin t - \cos t \ln(\sec t + \tan t)$
 $y = c_1 \cos(\ln x) + c_2 \sin(\ln x) - \cos(\ln x) \times \ln(\sec \ln x + \tan \ln x).$

Ex.1 $x^3 y'' - 3x^2 y' + 6xy' - 6y = x^4 \ln x.$

Ex.2 $2x^2 y'' + 4xy' + 3y = x + \ln x.$

Ex.3 $x^2 y'' + 8xy' + 4y = \sin(\ln x^2).$

Ex.3 $(x^3 D^3 - 3x^2 D^2 + 6xD - 6)y = x^4 \ln x.$

Sol: Put $x = e^t$, $x D = D_1$, $x^2 D^2 = D_1(D_1 - 1)$
 $x^3 D^3 = D_1(D_1 - 1)(D_1 - 2)$, $D_1 = \frac{d}{dt}$

$$\therefore (D_1(D_1 - 1)(D_1 - 2) - 3D_1(D_1 - 1) + 6D_1 - 6)y = e^{4t} \cdot t$$

$$\Rightarrow (D_1 - 1)(D_1(D_1 - 2) - 3D_1 + 6)y = t e^{4t}.$$

$$\Rightarrow (D_1 - 1)(D_1 - 2)(D_1 - 3)y = t e^{4t}.$$

$\therefore$ Put $y = e^{mt}$ in $(D_1 - 1)(D_1 - 2)(D_1 - 3)y = 0.$
 $\Rightarrow (m - 1)(m - 2)(m - 3) = 0$, $m = 1, 2, 3.$

$\therefore$ C.F = $c_1 e^t + c_2 e^{2t} + c_3 e^{3t}.$
 $= c_1 x + c_2 x^2 + c_3 x^3 \quad (\because x = e^t)$

$$\text{PI} = \frac{1}{(D_1 - 1)(D_1 - 2)(D_1 - 3)} t e^{4t}$$

$$= e^{4t} \frac{1}{(D_1 + 4 - 1)(D_1 + 4 - 2)(D_1 + 4 - 3)} t$$

$$\begin{aligned}
&= e^{4t} \frac{1}{(D_1+3)(D_1+2)(D_1+1)} t \\
&= e^{4t} \frac{1}{(D_1+3)(D_1+2)} (1+D_1)^{-1} t \\
&= e^{4t} \frac{1}{(D_1+3)(D_1+2)} (1-D_1+D_1^2-\dots) t \\
&= e^{4t} \frac{1}{(D_1+3)(D_1+2)} (t-1+0) \\
&= \frac{e^{4t}}{2} \frac{1}{(D_1+3)} \left(1+\frac{D_1}{2}\right)^{-1} (t-1) \\
&= \frac{e^{4t}}{2} \frac{1}{(D_1+3)} \left(1-\frac{D_1}{2}+\frac{D_1^2}{2^2}-\dots\right) (t-1) \\
&= \frac{e^{4t}}{2} \frac{1}{(D_1+3)} \left(t-1-\frac{1}{2}+0\right) \\
&= \frac{e^{4t}}{6} \left(1+\frac{D_1}{3}\right)^{-1} (t-3/2) \\
&= \frac{e^{4t}}{6} \left(1-\frac{D_1}{3}+\frac{D_1^2}{3^2}-\dots\right) (t-3/2) \\
&= \frac{e^{4t}}{6} \left(t-3/2-1/3\right) \\
&= \frac{e^{4t}}{6} \left(t-\frac{11}{6}\right) \\
&= \frac{x^4}{6} \left(\ln x - \frac{11}{6}\right).
\end{aligned}$$

General sol $y = C.F. + P.I$

$$= c_1 x + c_2 x^2 + c_3 x^3 + \frac{x^4}{6} \left(\ln x - \frac{11}{6}\right).$$

Ex $(x^4 D^4 - 24) y = x^5.$
Sol put $x = e^t.$

$$(x^4 D^4 - 24) y = (D_1(D_1-1)(D_1-2)(D_1-3) - 24) y = e^{5t}.$$

$$\Rightarrow (D_1^4 - 6D_1^3 + 11D_1^2 - 6D_1 - 24) y = e^{5t}$$

$$(D_1^4 + D_1^3 - 7D_1^3 - 7D_1^3 + 18D_1^2 + 18D_1 - 24D_1 - 24) y = e^{5t}$$

$$\Rightarrow (D_1+1) (D_1^3 - 7D_1^2 + 18D_1 - 24) y = e^{5t}$$

$$(D_1+1) (D_1^3 - 4D_1^2 - 3D_1^2 + 12D_1 + 6D_1 - 24) y = e^{5t}$$

$$(D_1+1) (D_1-4) (D_1^2 - 3D_1 + 6) y = e^{5t}$$

Put $y = e^{mt}$ in $(D_1+1) (D_1-4) (D_1^2 - 3D_1 + 6) = 0$

$$\Rightarrow (m+1)(m-4)(m^2 - 3m + 6) = 0.$$

$$\Rightarrow m = -1, 4, \frac{3 \pm \sqrt{9-24}}{2} = \frac{3 \pm i\sqrt{15}}{2}$$

$$\therefore \text{C.F.} = c_1 e^{-t} + c_2 e^{4t} + c_3 e^{\frac{(3+i\sqrt{15})}{2}t} + c_4 e^{\frac{(3-i\sqrt{15})}{2}t}$$

$$\text{or C.F.} = c_1 e^{-t} + c_2 e^{4t} + c_3 e^{\frac{3}{2}t} \cos \frac{\sqrt{15}}{2}t$$

$$+ c_4 e^{\frac{3}{2}t} \sin \frac{\sqrt{15}}{2}t.$$

$$= c_1 \frac{1}{x} + c_2 x^4 + c_3 x^{\frac{3}{2}} \cos\left(\frac{\sqrt{15}}{2} \ln x\right) + c_4 x^{\frac{3}{2}} \sin\left(\frac{\sqrt{15}}{2} \ln x\right).$$

$$\text{P.I.} = \frac{1}{(D_1+1)(D_1-4)(D_1^2-3D_1+6)} e^{5t}$$

$$= \frac{1}{(5+1)(5-4)(5^2-3 \times 5+6)} e^{5t}$$

$$= \frac{1}{6 \times 1 \times 16} = \frac{1}{96} e^{5t} = \frac{x^5}{96}.$$

General sol $y = \text{C.F.} + \text{P.I.}$